

Analysis First-Year Exam, August 2018, Part 1

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $(a_n)_{n=0}^{\infty}$ be a sequence of positive real numbers. Prove that

$$1/\limsup a_n = \liminf(1/a_n)$$

in the sense that if one side lies in $(0, \infty)$ then so does the other and the two sides are equal.

2. Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers and $A = \{a_n : n \geq 0\}$ its set of values. For p a real number, consider the two implications. One of these is true and the other false; give a proof for the true and a counterexample for the false.

- (i) if p is a limit point of A then p is the limit of a subsequence of $(a_n)_{n=0}^{\infty}$;
- (ii) if p is the limit of a subsequence of $(a_n)_{n=0}^{\infty}$ then p is a limit point of A .

3. Let X be a compact metric space and let $f : X \rightarrow X$ be a continuous function. Prove that if f has no fixed points, then there exists $a > 0$ such that $d(x, f(x)) \geq a$ whenever $x \in X$. Show by example that the compact hypothesis cannot be dropped.

4. This problem has two parts.

- (i) Prove that if c is any real number then the following subset of the Euclidean plane is connected

$$\{(x, y) : (x = \pm 1) \text{ or } (|x| < 1 \text{ and } y \geq c)\}.$$

- (ii) Let $C_0 \supseteq C_1 \supseteq \dots$ be a decreasing sequence of closed, connected subsets of a metric space. If the intersection $\bigcap_{n \geq 0} C_n$ is nonempty, must it be connected? Proof or counterexample.

5. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be differentiable and assume that $f'(t) \rightarrow 0$ as the real number t tends to infinity. Show that if $f(n) \rightarrow \ell$ as the integer n tends to infinity, then $f(t) \rightarrow \ell$ as the real number t tends to infinity.